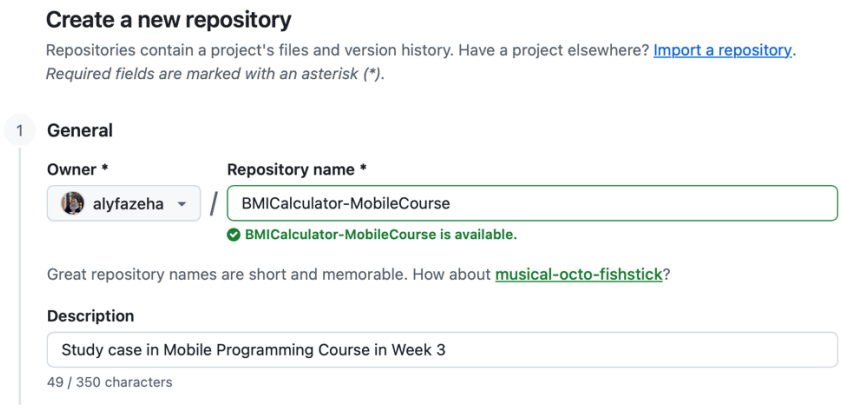
Name : **Alyfa Zahra Qurrota Aini**
NIM : **244107060057**
Major : **Business Infromation System**
Class : **2G**

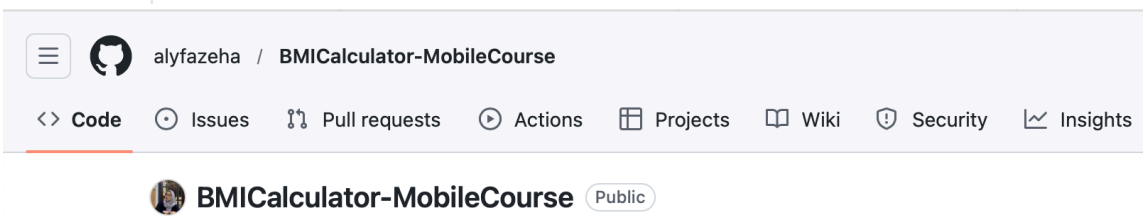
JOBSHEET 3
Percabangan, Perulangan, dan Operasi Git Lanjutan pada Dart

Studi Kasus: Aplikasi Kalkulator BMI

Alur Kerja Git:

1. Buat repository baru di GitHub





2. Clone repository ke local

```
Mobile — -zsh — 80×24
Last login: Sun Mar  8 20:21:45 on console
mac@alyfa-zahras-MacBook-Pro ~ % cd /Users/mac/Documents/KULIAH/SEMESTER\ 4/Mobile
mac@alyfa-zahras-MacBook-Pro Mobile % git clone https://github.com/alyfazeha/BMICalculator-MobileCourse
Cloning into 'BMICalculator-MobileCourse'...
warning: You appear to have cloned an empty repository.
mac@alyfa-zahras-MacBook-Pro Mobile %
```

3. Buat branch untuk setiap fitur:
   - branch input-validation

```
● mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %  git branch input-validation
● mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %  git branch
    input-validation
  * main
○ mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %
```

- branch bmi-calculation

```
● mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %  git branch bmi-calculation
● mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %  git branch
    bmi-calculation
    input-validation
  * main
○ mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %
```

- branch history-feature

```
● mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %  git branch history-feature
● mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %  git branch
    bmi-calculation
    history-feature
    input-validation
  * main
○ mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %
```

4. Merge branch ketika fitur selesai
   - Merge from branch input-validation

```
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git checkout input-validation
  Already on 'input-validation'
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git add .
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git commit -m "Adding the code program to validate the input
   cannot under 0"
  [input-validation 5aa1610] Adding the code program to validate the input cannot under 0
   1 file changed, 11 insertions(+), 1 deletion(-)
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git checkout main
  Switched to branch 'main'
  Your branch is based on 'origin/main', but the upstream is gone.
    (use "git branch --unset-upstream" to fixup)
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git merge input-validation
  Updating 7c8e34b..5aa1610
  Fast-forward
   src/bmi_calculator/bin/bmi_calculator.dart | 12 +++++++++++-
   1 file changed, 11 insertions(+), 1 deletion(-)
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % dart run
  Building package executable...
  Built bmi_calculator:bmi_calculator.
  Peringatan: Tinggi (155.0) dan Berat (-55.0) harus lebih dari 0.

  === Riwayat Perhitungan BMI ===
  Tinggi: 170.0cm | Berat: 65.0kg | BMI: 22.5 (Normal)
  Tinggi: 165.0cm | Berat: 75.0kg | BMI: 27.5 (Gemuk)
  Tinggi: 180.0cm | Berat: 85.0kg | BMI: 26.2 (Gemuk)
✧ mac@alyfa-zahras-MacBook-Pro bmi_calculator %
```

   - Merge from branch bmi-calculation

```
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git checkout bmi-calculation
  Switched to branch 'bmi-calculation'
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git add .
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git commit -m "Adding the code program to calculate BMI and
  show the category"
  [bmi-calculation d1f1685] Adding the code program to calculate BMI and show the category
   9 files changed, 488 insertions(+)
   create mode 100644 src/bmi_calculator/.gitignore
   create mode 100644 src/bmi_calculator/CHANGELOG.md
   create mode 100644 src/bmi_calculator/README.md
   create mode 100644 src/bmi_calculator/analysis_options.yaml
   create mode 100644 src/bmi_calculator/bin/bmi_calculator.dart
   create mode 100644 src/bmi_calculator/lib/bmi_calculator.dart
   create mode 100644 src/bmi_calculator/pubspec.lock
   create mode 100644 src/bmi_calculator/pubspec.yaml
   create mode 100644 src/bmi_calculator/test/bmi_calculator_test.dart
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git checkout main
  Switched to branch 'main'
  Your branch is based on 'origin/main', but the upstream is gone.
    (use "git branch --unset-upstream" to fixup)
● mac@alyfa-zahras-MacBook-Pro bmi_calculator % git merge bmi-calculation
  Updating 6b11cb9..d1f1685
  Fast-forward
   src/bmi_calculator/.gitignore                      |   3 +
   src/bmi_calculator/CHANGELOG.md                    |   3 +
   src/bmi_calculator/README.md                       |   2 +
   src/bmi_calculator/analysis_options.yaml           |  30 ++++
   src/bmi_calculator/bin/bmi_calculator.dart         |  27 ++++
   src/bmi_calculator/lib/bmi_calculator.dart         |   3 +
```

- Merge from branch history-feature

```
mac@alyfa-zahras-MacBook-Pro bmi_calculator % git checkout history-feature
M       src/bmi_calculator/bin/bmi_calculator.dart
Already on 'history-feature'
mac@alyfa-zahras-MacBook-Pro bmi_calculator % git add .
mac@alyfa-zahras-MacBook-Pro bmi_calculator % git commit -m "Adding the code program to save the history a
nd print history"
[history-feature 7c8e34b] Adding the code program to save the history and print history
 1 file changed, 26 insertions(+), 4 deletions(-)
mac@alyfa-zahras-MacBook-Pro bmi_calculator % git checkout main
Switched to branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)
mac@alyfa-zahras-MacBook-Pro bmi_calculator % git merge history-feature
Updating d1f1685..7c8e34b
Fast-forward
 src/bmi_calculator/bin/bmi_calculator.dart | 30 ++++++++++++++++++++++++----
 1 file changed, 26 insertions(+), 4 deletions(-)
mac@alyfa-zahras-MacBook-Pro bmi_calculator % dart run
Building package executable...
Built bmi_calculator:bmi_calculator.
=== Riwayat Perhitungan BMI ===
Tinggi: 170.0cm | Berat: 65.0kg | BMI: 22.5 (Normal)
Tinggi: 165.0cm | Berat: 75.0kg | BMI: 27.5 (Gemuk)
Tinggi: 180.0cm | Berat: 85.0kg | BMI: 26.2 (Gemuk)
mac@alyfa-zahras-MacBook-Pro bmi_calculator %
```

5. Push ke repository remote

```
mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse % git add .
mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse % git commit -m "BMI Calculation Program Officiall
y Done and Add all the documentation"
[main 971561d] BMI Calculation Program Officially Done and Add all the documentation
 11 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 docs/1. Make a new repository(1).png
 create mode 100644 docs/10. Result of the program.png
 create mode 100644 docs/11. Pust all the code to repository.png
 create mode 100644 docs/2. Make a new repository(2).png
 create mode 100644 docs/3. Clone repository to local.png
 create mode 100644 docs/4. Create branch input-validation.png
 create mode 100644 docs/5. Create branch bmi-calculation.png
 create mode 100644 docs/6. Create branch history-feature.png
 create mode 100644 docs/7. Merge from input-validation branch to main branch.png
 create mode 100644 docs/8. Merge from bmi-calculation branch to main branch.png
 create mode 100644 docs/9. Merge from history-feature branch to main branch.png
mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse % git push origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 8 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (14/14), 1.29 MiB | 2.19 MiB/s, done.
Total 14 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/alyfazeha/BMICalculator-MobileCourse
   5aa1610..971561d  main -> main
mac@alyfa-zahras-MacBook-Pro BMICalculator-MobileCourse %
```

Fitur Aplikasi:

Input tinggi dan berat badan, Kalkulasi BMI, Kategorisasi hasil (kurus, normal, gemuk), dan Riwayat perhitungan (menggunakan perulangan)

```
mac@alyfa-zahras-MacBook-Pro bmi_calculator % dart run
Building package executable...
Built bmi_calculator:bmi_calculator.
Peringatan: Tinggi (155.0) dan Berat (-55.0) harus lebih dari 0.

=== Riwayat Perhitungan BMI ===
Tinggi: 170.0cm | Berat: 65.0kg | BMI: 22.5 (Normal)
Tinggi: 165.0cm | Berat: 75.0kg | BMI: 27.5 (Gemuk)
Tinggi: 180.0cm | Berat: 85.0kg | BMI: 26.2 (Gemuk)
mac@alyfa-zahras-MacBook-Pro bmi_calculator %
```